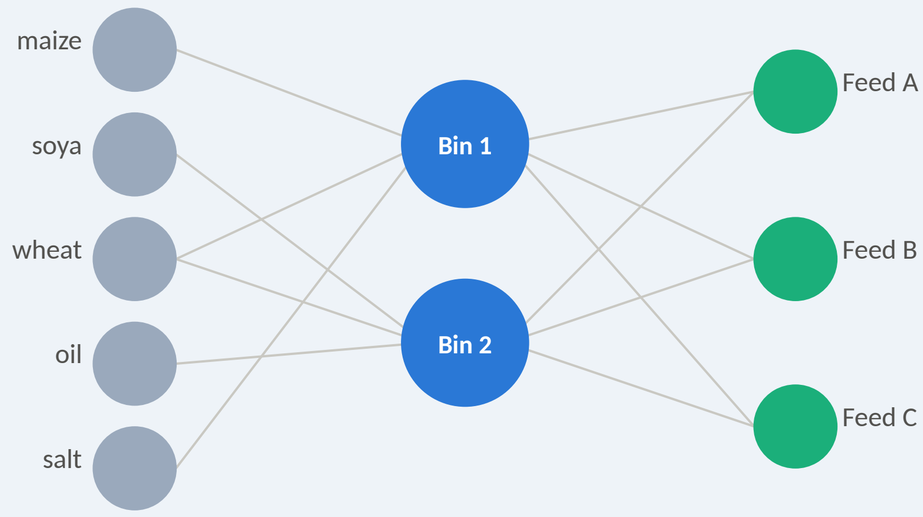
Smarter Restarts for Feed-Blending Optimization

Improving Sequential Linear Programming (SLP) for the pooling problem: from our IA-1 re-implementation to IA-2

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Introduction to Optimization Techniques (IA-2) · Dept. of IT, K. J. Somaiya College of Engineering

1 The problem



Raw materials go through a few **shared bins** into many feeds. Every feed must meet nutrient limits. **Goal: the lowest total cost.**

2 Why it is hard, and the gap

A bin that suits one feed can hurt another: quality = share × content (two unknowns multiplied), so the problem has many **'valleys'** (local optima).

The paper's SLP (our IA-1 code) restarts from **random** points, so it often ends in a poor valley.

6%

Good runs on the 6 hard instances (IA-1)

12%

of all IA-1 runs end infeasible

3 Our idea (IA-2): restart smart, not random

1 — Smart start

Compute each feed's ideal recipe (a small LP) and group similar recipes into the bins.

2 — Fix one, solve the other

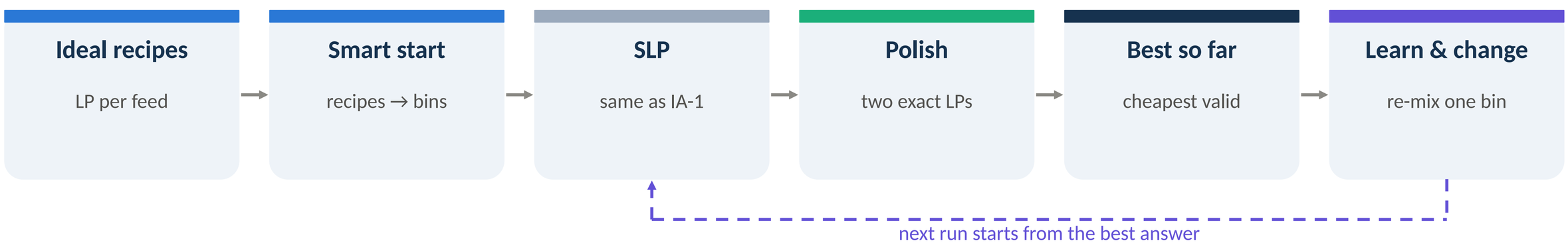
With the bins fixed the rest is an easy LP: it repairs and polishes every answer.

3 — Learn from the best

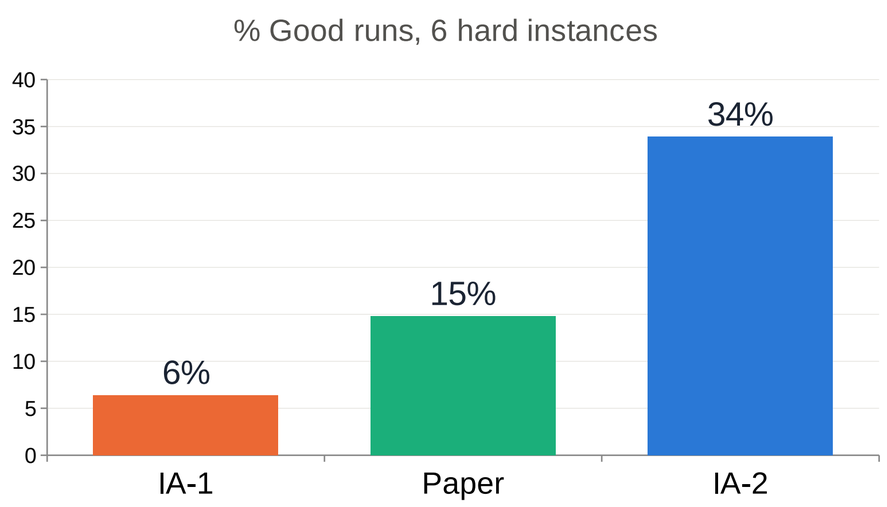
Each new run starts from the best answer so far and changes the bin that hurts one feed most.

The model and the SLP solver from IA-1 stay unchanged. Check on af-1 (our IA-1 instance): IA-1 and IA-2 are both Good in 100% of runs, and IA-2's lower bound (295.93 = BARON's) proves the cost 296.446 is within 0.17% of the optimum.

4 How IA-2 runs (20 runs per instance)



5 Results: IA-1 vs IA-2 (11 instances, 3 × 20 runs each, same computer)



5×

faster to a Good answer (hard)

88→96%

feasible runs (all 11)

44→62%

Good runs (all 11)

Biggest gains: af-2 2% → 68%, af-4 20% → 58% Good runs. Limits: af-6-f unsolved by both; easy instances a bit slower.

Take-away: using simple LPs to start smart and to learn from the best answer makes SLP find good feed recipes about 5× more often on hard problems.

[1] A. Grothey, K. McKinnon, On the effectiveness of sequential linear programming for the pooling problem, Ann. Oper. Res. 322 (2023) 691–711. [2] Group OPT-09, IA-1 report (2026). Code + Colab demo: github.com/razzantambe/OPT-09-IA2-Pooling-SLP